Лаб 4

Подготовка:

```
PS D:\ITMO\DataEng\zookeeper-failover> docker compose up -d --wait zk1 zk2 zk3 receiver
[+] up 4/4
 ✔Container zookeeper-failover-zk1-1      Healthy
 ✔Container zookeeper-failover-zk2-1      Healthy
 ✔Container zookeeper-failover-zk3-1      Healthy
 ✔Container zookeeper-failover-receiver-1 Healthy
```

```
PS D:\ITMO\DataEng\zookeeper-failover> docker compose ps
NAME                          IMAGE                          COMMAND                  SERVICE    CREATED              STATUS
PORTS
zookeeper-failover-receiver-1   zookeeper-failover-app:local   "python src/receiver…"   receiver   About a minute ago   Up About a minute (healthy)
zookeeper-failover-zk1-1        zookeeper:3.9.6                "/docker-entrypoint.…"   zk1        About a minute ago   Up About a minute (healthy)
127.0.0.1:2181->2181/tcp
zookeeper-failover-zk2-1        zookeeper:3.9.6                "/docker-entrypoint.…"   zk2        About a minute ago   Up About a minute (healthy)
127.0.0.1:2182->2181/tcp
zookeeper-failover-zk3-1        zookeeper:3.9.6                "/docker-entrypoint.…"   zk3        About a minute ago   Up About a minute (healthy)
127.0.0.1:2183->2181/tcp
PS D:\ITMO\DataEng\zookeeper-failover> docker compose exec -T zk1 java -version
openjdk version "17.0.21" 2026-10-20
OpenJDK Runtime Environment Temurin-17.0.21+9 (build 17.0.21+9)
OpenJDK 64-Bit Server VM Temurin-17.0.21+9 (build 17.0.21+9, mixed mode, sharing)
PS D:\ITMO\DataEng\zookeeper-failover> docker compose exec -T zk1 cat /conf/zoo.cfg
dataDir=/data
dataLogDir=/datalog
tickTime=2000
initLimit=10
syncLimit=5
autopurge.snapRetainCount=3
autopurge.purgeInterval=0
maxClientCnxns=60
standaloneEnabled=true
admin.enableServer=false
server.1=zk1:2888:3888;2181
server.2=zk2:2888:3888;2181
server.3=zk3:2888:3888;2181
4lw.commands.whitelist=ruok,srvr,mntr,stat
PS D:\ITMO\DataEng\zookeeper-failover>
```

```
zk1
Mode: follower
zk2
Mode: leader
zk3
Mode: follower
```

Остановка одного контейнера:

```
PS D:\ITMO\DataEng\zookeeper-failover> docker compose stop zk1
[+] stop 1/1
 ✔Container zookeeper-failover-zk1-1 Stopped                                                                                0.7s
PS D:\ITMO\DataEng\zookeeper-failover> docker compose ps -a
NAME                              IMAGE                          COMMAND                  SERVICE         CREATED          STATUS
          PORTS
zookeeper-failover-coordinator-a-1   zookeeper-failover-app:local   "python src/coordina…"   coordinator-a   55 minutes ago   Exited (0) 54 minutes ag
o
zookeeper-failover-coordinator-b-1   zookeeper-failover-app:local   "python src/coordina…"   coordinator-b   55 minutes ago   Exited (0) 54 minutes ag
o
zookeeper-failover-receiver-1        zookeeper-failover-app:local   "python src/receiver…"   receiver        3 minutes ago    Up 3 minutes (healthy)

zookeeper-failover-zk1-1             zookeeper:3.9.6                "/docker-entrypoint.…"   zk1             3 minutes ago    Exited (143) Less than a
 second ago
zookeeper-failover-zk2-1             zookeeper:3.9.6                "/docker-entrypoint.…"   zk2             3 minutes ago    Up 3 minutes (healthy)
          127.0.0.1:2182->2181/tcp
zookeeper-failover-zk3-1             zookeeper:3.9.6                "/docker-entrypoint.…"   zk3             3 minutes ago    Up 3 minutes (healthy)
          127.0.0.1:2183->2181/tcp
PS D:\ITMO\DataEng\zookeeper-failover> docker compose exec -T receiver python src/probe.py write --hosts zk2:2181
{"available": true, "write_read_delete": "quorum-write-ok"}
PS D:\ITMO\DataEng\zookeeper-failover>
```

Остановка еще одного:

```
PS D:\ITMO\DataEng\zookeeper-failover> docker compose ps -a
NAME                                IMAGE                          COMMAND                  SERVICE         CREATED          STATUS
      PORTS
zookeeper-failover-coordinator-a-1   zookeeper-failover-app:local   "python src/coordina…"   coordinator-a   56 minutes ago   Exited (0) 55 minutes ag
o
zookeeper-failover-coordinator-b-1   zookeeper-failover-app:local   "python src/coordina…"   coordinator-b   56 minutes ago   Exited (0) 55 minutes ag
o
zookeeper-failover-receiver-1        zookeeper-failover-app:local   "python src/receiver…"   receiver        4 minutes ago    Up 3 minutes (healthy)

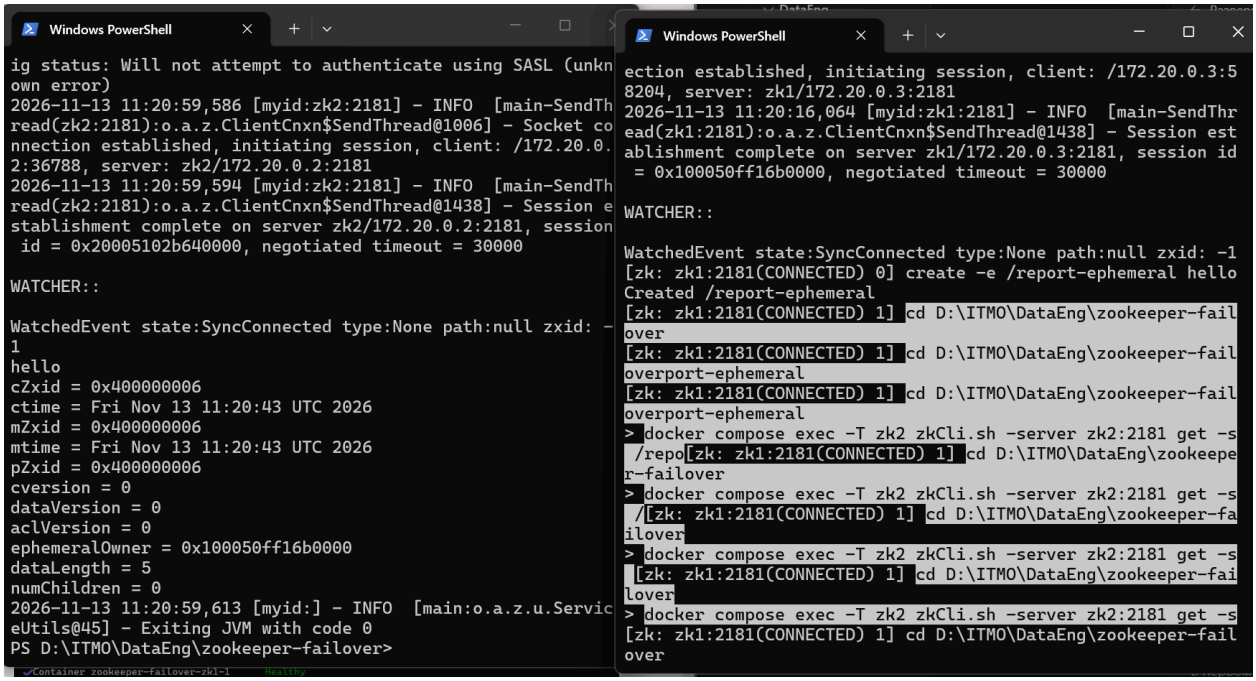
zookeeper-failover-zk1-1             zookeeper:3.9.6                "/docker-entrypoint.…"   zk1             4 minutes ago    Exited (143) 56 seconds
ago
zookeeper-failover-zk2-1             zookeeper:3.9.6                "/docker-entrypoint.…"   zk2             4 minutes ago    Up 4 minutes (healthy)
      127.0.0.1:2182->2181/tcp
zookeeper-failover-zk3-1             zookeeper:3.9.6                "/docker-entrypoint.…"   zk3             4 minutes ago    Exited (143) 5 seconds a
go
PS D:\ITMO\DataEng\zookeeper-failover>
PS D:\ITMO\DataEng\zookeeper-failover> docker compose exec -T receiver python src/probe.py write --hosts zk2:2181 --expect-unavailable
Connection dropped: socket connection broken
Connection dropped: socket connection broken
Connection dropped: socket connection broken
Connection dropped: socket connection broken
Connection dropped: socket connection broken
Connection dropped: socket connection broken
Connection dropped: socket connection broken
Failed connecting to Zookeeper within the connection retry policy.
{"available": false, "error": "KazooTimeoutError"}
PS D:\ITMO\DataEng\zookeeper-failover>
```

Восстановление:

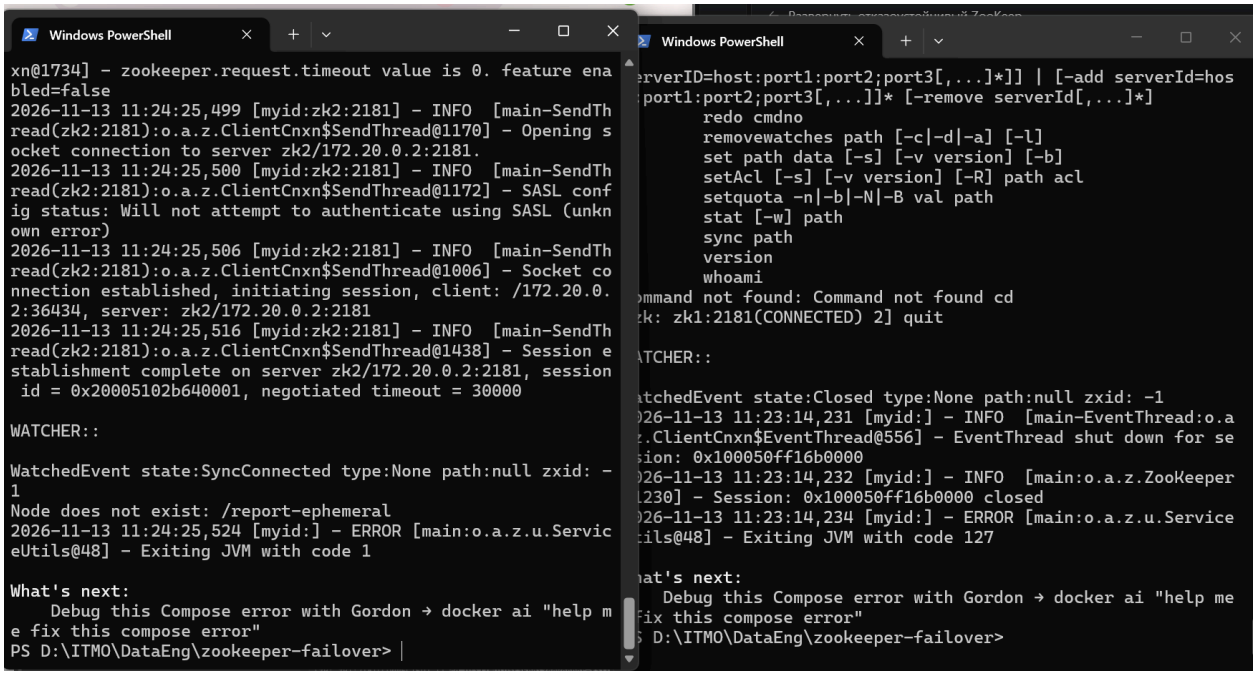
```
{"available": false, "error": "KazooTimeoutError"}
PS D:\ITMO\DataEng\zookeeper-failover> docker compose start zk1 zk3
[+] start 2/2
 ✔Container zookeeper-failover-zk1-1 Started
 ✔Container zookeeper-failover-zk3-1 Started
PS D:\ITMO\DataEng\zookeeper-failover> docker compose up -d --wait zk1 zk2 zk3 receiver
[+] up 4/4
 ✔Container zookeeper-failover-zk2-1      Healthy
 ✔Container zookeeper-failover-zk3-1      Healthy
 ✔Container zookeeper-failover-receiver-1 Healthy
 ✔Container zookeeper-failover-zk1-1      Healthy
PS D:\ITMO\DataEng\zookeeper-failover>
PS D:\ITMO\DataEng\zookeeper-failover> docker compose exec -T receiver python src/probe.py write
{"available": true, "write_read_delete": "quorum-write-ok"}
PS D:\ITMO\DataEng\zookeeper-failover> docker compose exec -T receiver python src/probe.py roles |
>>     Select-String '^zk[123]|Mode:|unavailable|not currently serving'

zk1
Mode: follower
zk2
This ZooKeeper instance is not currently serving requests
zk3
Mode: leader
```

Создание эфемерного узла:



Закрываем сессию, проверяем исчезновение узла:



Запуск координаторов и приостановка лидеров:

actor	event	session_timeout
coordinator-a	candidate	10
coordinator-b	candidate	10

event	epoch	rows
became_leader	1	
checkpoint_saved	200	
checkpoint_saved	400	
checkpoint_saved	600	

```
> logs\commands.txt:389:$ docker compose pause coordinator-b
  logs\commands.txt:390: Container zookeeper-failover-coordi
nator-b-1 Paused
  logs\commands.txt:391:
  logs\commands.txt:392:exit_code=0
```

utc	event	epoch
2026-11-13T10:20:20.347241+00:00	became_leader	2
2026-11-13T10:20:20.370802+00:00	restored_checkpoint	
2026-11-13T10:20:42.611784+00:00	publication_accepted	2

Две публикации без защиты:

```
> Format-Table coordinator,epoch,message,pointer_version
-AutoSize

oordinator   epoch message           pointer_version
----------   ----- -------           ---------------
oordinator-a     2 записал результат               1
oordinator-b     1 записал результат               2


S D:\ITMO\DataEng\zookeeper-failover>
S D:\ITMO\DataEng\zookeeper-failover> $r.runs[0].final_point
r | ConvertTo-Json -Depth 5

   "value":  {
                "coordinator":  "coordinator-b",
                "epoch":  1,
                "fencing":  false,
                "pair_occurrences":  124557,
                "path":  "results/unfenced-20261113T101907Z
coordinator-b-epoch-1.csv",
                "rows":  6907,
                "sha256":  "939774ba3609a093ee33edb348e2528
14206c801f38dcb46d2d561e7e1b7847",
                "unique_pairs":  20635
            },
   "version":  2
```

Отклонение устаревшей записи с защитой:

```
event     operation   coordinator   epoch pointer_version er
                                                          ro
                                                          r
-----     ---------   -----------   ----- --------------- --
published             coordinator-a     2               1
rejected  /checkpoint coordinator-b     1                 Ba
rejected  /publish    coordinator-b     1                 Ba


PS D:\ITMO\DataEng\zookeeper-failover>
PS D:\ITMO\DataEng\zookeeper-failover> $r.runs[1].final_point
er | ConvertTo-Json -Depth 5
{
    "value":  {
                "coordinator":  "coordinator-a",
                "epoch":  2,
                "fencing":  true,
                "pair_occurrences":  124557,
                "path":  "results/fenced-20261113T101907Z/c
oordinator-a-epoch-2.csv",
                "rows":  6907,
                "sha256":  "939774ba3609a093ee33edb348e2528
414206c801f38dcb46d2d561e7e1b7847",
                "unique_pairs":  20635
            },
    "version":  1
}
```

Время смены лидера:

```
fencing restored_rows failover_seconds pause_seconds
------- ------------ ----------------- -------------
  False           600         8,570655     40,501706
   True           600        10,215975     40,431458
```

Совпадение с ДЗ 3:

```
PS D:\ITMO\DataEng\zookeeper-failover> python scripts\verify_
results.py
unfenced-20261113T101907Z: OK; publications=2; pointer_versio
n=2; restored_rows=600; failover=8.571s
fenced-20261113T101907Z: OK; publications=1; pointer_version=
1; restored_rows=600; failover=10.216s
All saved evidence and every pair count match homework 3.
PS D:\ITMO\DataEng\zookeeper-failover>
PS D:\ITMO\DataEng\zookeeper-failover> $r.runs[0].final_point
er.value |
>>     Format-List rows,unique_pairs,pair_occurrences


rows             : 6907
unique_pairs     : 20635
pair_occurrences : 124557
```

Почему три сервера:
Для работы требуется большинство. Из двух серверов нужны оба, поэтому отказ одного останавливает работу. Из трёх достаточно двух. Из четырёх нужны также три, расход ресурсов больше, а отказоустойчивость такая же. Поэтому 3 - оптимальное количество.

Почему эпоху проверяет получатель:
После потери сессии старый координатор может продолжить вычисления и отправить результат. ZooKeeper не может остановить этот процесс или запретить запись во внешнюю систему, поэтому и нужна проверка получателя